

— ML Performance (2026-08-01 s/d 2026-08-30)

Metric	Value
Evaluation Generated At (UTC)	2026-07-26T12:34:13.201368+00:00
=== 1. MODEL PERFORMANCE SUMMARY ===	
--- AGENUSA ---	
Active Model Version	agenusa_v20260801102513835592
Model Created At	2026-08-01 10:25:13
Dataset Size	5000
Anomaly Count (default)	250
Anomaly Rate	5.0%
Contamination Rate	0.05
Review Score Threshold	0.004
High-Risk Score Threshold	-0.0026
Predictions at Review Boundary	499
Numeric Features Count	17
Categorical Features Count	3
Numeric Features	AMOUNT, PROCESSING_CODE, RESPONSE_CODE, MTI, TX_HOUR, TX_DAYOFWEEK, IS_NIGHT_TX, GAP_MINUTES, TERMINAL_SWITCH_FAST, AVG_AMOUNT_5, AMOUNT_OVER_AVG_RATIO, IS_DECLINED, IS_BRUTE_PATTERN, IS_MONEY_MULE_DEST, IS_HIGH_AMOUNT_PATTERN, MIDNIGHT_AMOUNT_SPIKE, RAPID_RETRY_DECLINED
Categorical Features	TERMINAL_ID, MERCHANT_ID, DEST_ACCOUNT_NUMBER

--- NUSABILL ---	
Active Model Version	nusabill_v20260801102530041908
Model Created At	2026-08-01 10:25:29
Dataset Size	5000
Anomaly Count (default)	250
Anomaly Rate	5.0%
Contamination Rate	0.05
Review Score Threshold	0.0357
High-Risk Score Threshold	-0.0298
Predictions at Review Boundary	501
Numeric Features Count	12
Categorical Features Count	2
Numeric Features	BILL_AMOUNT, PAYMENT_AMOUNT, REFUND_FLAG, PAYMENT_DELAY_DAYS, PAYMENT_TO_BILL_RATIO, UNDERPAY_FLAG, HIGH_SPIKE_FLAG, CHANNEL_API_FLAG, PAYMENT_GAP_MINUTES, BURST_FLAG, CHANNEL_SWITCH_TO_API, EARLY_PAYMENT_ANOMALY
Categorical Features	CHANNEL, BILL_STATUS
=== 2. RETRAIN HISTORY ===	Periode: 2026-08-01 s/d 2026-08-30

Metric	Value
Total Retrain Executions	4
Successful Retrains	3
Retrain Success Rate (%)	75.0
Total Records Trained (cumulative)	0 (dari 0/4 eksekusi yang punya data ini)
Total Feedback Records Used (cumulative)	10 (dari 3/4 eksekusi yang punya data ini)
Total Anomalies Found (cumulative)	1399
Total New Patterns Discovered (cumulative)	0
Trigger Type Breakdown	
- manual	4
Recent Retrain Executions (max 10)	
Execution Time Trigger Status Records Feedback Used New Patterns	
2026-08-01 10:25:29 manual SUCCESS	records=n/a, feedback=3, anomalies=500, patterns=0
2026-08-01 10:25:13 manual SUCCESS	records=n/a, feedback=4, anomalies=401, patterns=0
2026-08-01 10:22:04 manual FAILED	records=n/a, feedback=n/a, anomalies=0, patterns=0
2026-08-01 10:21:55 manual SUCCESS	records=n/a, feedback=3, anomalies=498, patterns=0
=== 3. PATTERN DISCOVERY ===	
Top 10 Patterns Category Risk Score Hit Count	
Agenusa - EDC Terminal Pooling & Card Washing Detection	category=NETWORK, risk_score=90, hits=32
Agenusa - Velocity Burst (High-Frequency Card Usage)	category=VELOCITY, risk_score=85, hits=31
AI Discovery: Fast Terminal Switch	category=LOCATION, risk_score=80, hits=28
Agenusa - Account Takeover & Brute Force PIN Guessing	category=CREDENTIAL, risk_score=75, hits=27
Auto Velocity AGENUSA (5+ tx in 5m)	category=VELOCITY, risk_score=40, hits=22
Agenusa - Critical Card Testing Burst Detection	category=VELOCITY, risk_score=95, hits=20
Nusabill - Fan-Out Spam (Fake Invoice Mass Blast)	category=NETWORK, risk_score=80, hits=9
Nusabill - Tuned Fake Invoice Blast Detection (Low Nominal Trap)	category=NETWORK, risk_score=85, hits=8
Agenusa - Super Pattern (Advanced Syndicate Attack)	category=COMPOSITE, risk_score=100, hits=4
Agenusa - Fan-In Syndicate (Massive Card Pooling)	category=NETWORK, risk_score=100, hits=4
Total Patterns in System	18
Active Patterns	10
=== 4. FRAUD DETECTION EFFECTIVENESS ===	Periode: 2026-08-01 s/d 2026-08-30
--- AGENUSA ---	
Total Transactions Processed	51
Actual Fraud Count	26
Actual Fraud Rate (%)	50.98
Model Anomaly Rate (%)	5.0
Anomaly Rate vs Actual Fraud Rate Gap	-45.98 pp (model kurang sensitif)

Metric	Value
---	NUSABILL ---
Total Transactions Processed	24
Actual Fraud Count	5
Actual Fraud Rate (%)	20.83
Model Anomaly Rate (%)	5.0
Anomaly Rate vs Actual Fraud Rate Gap	-15.83 pp (model kurang sensitif)
Total New Patterns Generated (all retrains, periode ini)	0
===	CATATAN ===
Akurasi Model	Tidak tersedia — dataset training bersifat unlabeled (tanpa kolom IS_FRAUD), sehingga metrik Precision, Recall, dan F1-Score tidak dapat dihitung.
Metode Evaluasi	Berbasis threshold boundary dari hasil training (Isolation Forest), bukan validasi terhadap ground truth.
Anomaly Rate vs Fraud Rate	Perbandingan ini bersifat indikatif — anomaly rate adalah parameter model (contamination), sedangkan fraud rate adalah hasil keputusan final (rule engine + pattern + ML + review analis).